# Muhammad Nur Shahril Iskandar

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## Education

**Griffith University** — Australia
*Australian Centre for Precision Health and Technology (PRECISE), School of Allied Health, Sport and Social Work* — Present
- Doctor of Philosophy (PhD) in Biomechanics
- Research focus: Personalised biomechanics enabled by precision technology for treatment of hip osteoarthritis

**Nanyang Technological University (NTU)** — Singapore
*Department: Physical Education and Sports Science Academic Group* — Jun 2023
- B.S., Sport Science and Management with Honours (Highest Distinction)

## Research Experience

**Senior Research Assistant, Nanyang Technological University** — Singapore
*Sports Biomechanics Lab | Principal Investigators: Phillis Teng, PhD & Kong Pui Wah, PhD* — Aug 2023 - Feb 2025
- Assisted in 2 main projects; 1) Screening and Biomechanical Risk Factors for Early Knee Osteoarthritis, 2) Coaching table tennis to individuals with physical disabilities.
- Contributed to the development of the research design and ethics application.
- Conducted a systematic review on the use of ultrasonography to diagnose early knee osteoarthritis using Covidence.
- Mentorred and coordinated with 2 undergraduate students in data collection in utilizing various equipment such as plantar pressure mapping, tensiomyography and thermal camera.
- Investigated the use of monocular camera to obtain biomechanical data in table tennis through markerless motion capture.
- Published a first-author conference paper on the use of markerless motion capture system in running [C3].

**Research Intern, Agency for Science, Technology and Research, Bioinformatics Institute** — Singapore
*Biophysical Modelling Lab | Principal Investigator: Chiam Keng Hwee, PhD* — Jan 2023 - Jun 2023
- Contributed to a project on data-driven gait rehabilitation of lower limb amputees.
- Implemented generative AI using Stable Diffusion to enhance existing open-source pose estimation algorithm (OpenPose & DeepLabCut) in identifying lower limb amputee's anatomical landmarks for gait analysis [J5].
- Involved in sourcing of walking videos, labeling of anatomical landmarks and training a model using DeepLapCut.
- Wrote a Python script to perform data processing and analysis.

**Undergraduate Research Assistant, Nanyang Technological University** — Singapore
*Sports Biomechanics Lab | Principal Investigator: Kong Pui Wah, PhD* — Apr 2022 - May 2023
- Contributed to a project analyzing the biomechanical effects of exoskeletal in military personnel.
- Assisted in the synchronized gait data collections using VICON Nexus on Bertec split-belt instrumented treadmill, Delsys EMG system and loadsol® sensors.
- Wrote MATLAB and Python scripts to extract data and conduct data analysis of ground reaction forces using statistical parametric mapping, spm1d.
- Publications: 2 conference abstract [CA2, CA3] and 2 journal articles [J2, J4].

**Undergraduate Research Programme (URECA), Nanyang Technological University** — Singapore
*Sports Biomechanics Lab | Principal Investigator: Kong Pui Wah, PhD* — Aug 2021 - Aug 2022
- Contributed to the development of a video-based analysis model for assessing treadmill running biomechanics.
- Facilitated over 40 participants' recruitment and utilized Kinovea to analyze running kinematics.
- Conferred the title "NTU President Research Scholar" for completing the programme with Distinction.
- Publications: 2 conference papers [C1, C2], 2 journal articles [J1, J3].

**Undergraduate Research Assistant, Nanyang Technological University** — Singapore
*Human Bioenergetics Lab | Principal Investigator: Yang Yifan, PhD* — Sep 2020 - Mar 2021
- Contributed to the project assessing the dose-response of leucine on muscle maintenance during weight loss.
- Independently recruited over 20 participants and coordinated weekly anthropometric measurements.
- Verified accuracy of participant's data entry for daily physical activity, sleep, and dietary intake log.

## Publications

* indicates corresponding author, † indicates equal contribution

### Peer-reviewed Journal Articles

[J8] **Iskandar, M. N. S.**, Teng, P.*, Mukherjee, S. & Tan, S. Y. (Accepted). Utility of ultrasound in the diagnosis of early knee osteoarthritis: A systematic review. *Sports Medicine and Arthroscopy Review*.

[J7] Loh, R. B. C, Pan, J. W., **Iskandar, M. N. S.**, & Kong, P. W.*(2026). Predicting running-related injuries from functional, kinetic and kinematic data. *International Journal of Sport Medicine*. 47(05):372-380. doi: 10.1055/a-2706-5516

[J6] Marican, M. A., Chandra, L., Tang, Y., **Iskandar, M. N. S.**, Lim, C. X. E. & Kong, P. W.*(2025). Biomechanical Effects of a Passive Back-Support Exosuit during Simulated Military Lifting Tasks - An EMG Study. *Sensors*. 25(10), 3211. doi: 10.3390/s25103211

[J5] Zhou, T†, **Iskandar, M. N. S.**†, & Chiam, K. H.* (2025). Diffusion Models Enable Zero-Shot Pose Estimation for Lower-Limb Prosthetic Users. *PLOS Digital Health*. 4(3), e0000745. doi: 10.1371/journal.pdig.0000745

[J4] Kong, P. W.*, Koh, A. H., Ho, M. Y. M., **Iskandar, M. N. S.**, & Lim, C. X. E. (2024). Effectiveness of A Passive Military Exoskeleton in Offloading Weight during Static and Dynamic Load Carriage: A Randomised Cross-Over Study. *Applied Ergonomics*. 119, 104293. doi: 10.1016/j.apergo.2024.104293

[J3] **Iskandar, M. N. S.**, Loh, R. B. C., Ho, M. Y. M., Pan, J. W. & Kong, P. W.* (2023). Crossover Gait in Running and Measuring Foot Inversion Angle at Initial Foot Strike: A Front-View Video Analysis Approach. *Frontiers in Bioengineering and Biotechnology*. 11, 1210049. doi: 10.3389/fbioe.2023.1210049

[J2] Kong, P. W.*, **Iskandar, M. N. S.**, Koh, A. H., Ho, M. Y. M., & Lim, C. X. E. (2023). Validation of In-Shoe Force Sensors During Loaded Walking In Military Personnel. *Sensors*. 23(14), 6465. doi: 10.3390/s23146465

[J1] Pan, J. W., Ho, M. Y. M., Loh, R. B. C., **Iskandar, M. N. S.**, & Kong, P. W.* (2023). Foot Morphology and Running Gait Pattern between the Left and Right Limbs in Recreational Runners. *Physical Activity and Health*, 7(1), 43–52. doi: 10.5334/paah.226

### Peer-reviewed Conference Proceedings

[C3] **Iskandar, M. N. S.**, & Teng, P. S. P.* (2024). Kinematics Comparison of OpenCap and IMU with Marker-Based Motion Capture In Treadmill Running: A Pilot Study. *Proceedings of the 42$^{nd}$ International Society of Biomechanics in Sports Conference.*

[C2] Loh, R. B. C., Ho, M. Y. M., **Iskandar, M. N. S.**, & Kong, P. W.* (2024). Two-Dimensional Kinematics Differences Between Sexes In Runners With and Without Patellofemoral Pain. *Proceedings of the 42$^{nd}$ International Society of Biomechanics in Sports Conference.*

[C1] **Iskandar, M. N. S.**, Loh, R. B. C., Ho, M. Y. M., Pan, J. W., & Kong, P. W.* (2022). Comparison of Rearfoot Inversion Angle at Initial Footstrike Measured From Front And Back View Videos. *Proceedings of the 40$^{th}$ International Society of Biomechanics in Sports Conference*, 40(1), 291.

### Conference Abstracts

[CA5] Teng, P. S. P., Mukherjee, S., Tan, S. Y., **Iskandar, M. N. S.**, Lim, M. Y. K., See, R. Y. & Lim, B. W. B. (2025). Plantar Pressure Differences Between Low and High-Risk Knee Osteoarthritis Long-Distance Recreational Runners. *30$^{th}$ International Society of Biomechanics Conference. July 2025.*

[CA4] **Iskandar, M. N. S.**, Quek, B. T. L., Liu, H., Ma, C. M. S., & Kong, P. W. Feasibility of Markerless Motion Capture in Non-Standardised Conditions - A Case Study on a Table Tennis Player with Physical Disabilities. *10$^{th}$ Asian Society of Sport Biomechanics Conference*. December 2024.

[CA3] Lim, C., Kong, P. W., Koh, A. H., Ho, M., & **Iskandar, M. N. S.**. The physiological and biomechanical effects of a full-body passive exoskeleton on military load carriage. *6th International Congress on Soldiers' Physical Performance*. September 2023.

[CA2] **Iskandar, M. N. S.**, Koh, A. H., Ho, M. Y. M., Lim, C. X. E., & Kong, P. W. Validation of the loadsol® in-shoe force sensors during walking in military boots under heavy load carriage. *9$^{th}$ Asian Society of Sport Biomechanics Conference*. August 2023.

[CA1] Loh, R. B. C., Ho, M. Y., **Iskandar, M. N. S.**, Pan, J. W., & Kong, P. W. Reliability of video-based running gait analysis in recreational runners. *XXII International Conference on Mechanics in Medicine and Biology*. August 2022.

## Conference Oral Presentations

[CP4] Feasibility of Markerless Motion Capture in Non-Standardised Conditions - A Case Study on a Table Tennis Player with Physical Disabilities, *$10^{th}$ Asian Society of Sports Biomechanics (ASSB) Conference*, Kuala Lumpur, Malaysia, December 2024

[CP3] Validation of the loadsol® in-shoe force sensors during walking in military boots under heavy load carriage, *$9^{th}$ Asian Society of Sports Biomechanics (ASSB) Conference*, Bangkok, Thailand, August 2023

[CP2] Two-Dimensional Video Analysis of the Rearfoot Inversion Angle at Initial Footstrike in Treadmill Running, *$10^{th}$ International Conference of Undergraduate Research (ICUR)*, Virtual presentation, September 2022

[CP1] Comparison of Rearfoot Inversion Angle at Initial Footstrike Measured From Front And Back View Videos, *$40^{th}$ International Society of Biomechanics in Sports (ISBS) Conference*, Liverpool, United Kingdom, July 2022

## Teaching Experience

Tutor - Introductory Biomechanics (2011AHS) 2026

## Community Service

Student Mentor 2025

*Queensland Academies for Health Sciences and Griffith University*

Mentored Year 10 students in developing innovative solutions to healthcare challenges through the nine-week Change in Action program, providing research guidance and feedback.

## Awards

Anugerah Cemerlang MENDAKI Award 2023

*Awarded by Yayasan MENDAKI to undergraduate Malay students for graduating with first-class honors/highest distinction.*

Internship Commendation Award 2023

*Awarded by NTU to the top 15% of the cohort for excellent work performed during the undergraduate internship.*

Best Thesis Oral Presentation Award 2022

*Awarded by NTU at the $11^{th}$ Lau Teng Chuan Physical Education & Sports Science Symposium.*

ISBS Student Travel Grant 2022

*Awarded by ISBS to attend the $40^{th}$ ISBS Conference.*

## Skills

**Programming:** Python, MATLAB, R, Excel VBA, HTML/CSS

**Software:** SPSS, JASP, OpenSim, Visual3D, VICON Nexus

**Languages:** English (Native), Malay